

Enterprise Infrastructure Planning for a Startup Company

Course: ITEP 414 – System Administration and Maintenance

Project: Week 2 Portfolio Project

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BSIT- 4B

PART 1 — COMPANY PROFILE

Company Name: ABC Startup Solutions

Nature of Business

ABC Startup Solutions is a startup software development company that provides customized web applications, business software, mobile applications, and technical solutions for small and medium-sized businesses. The company focuses on developing reliable and user-friendly digital solutions that help organizations improve their daily operations.

Company Vision

To become a trusted technology partner for businesses by providing innovative, secure, and reliable software solutions.

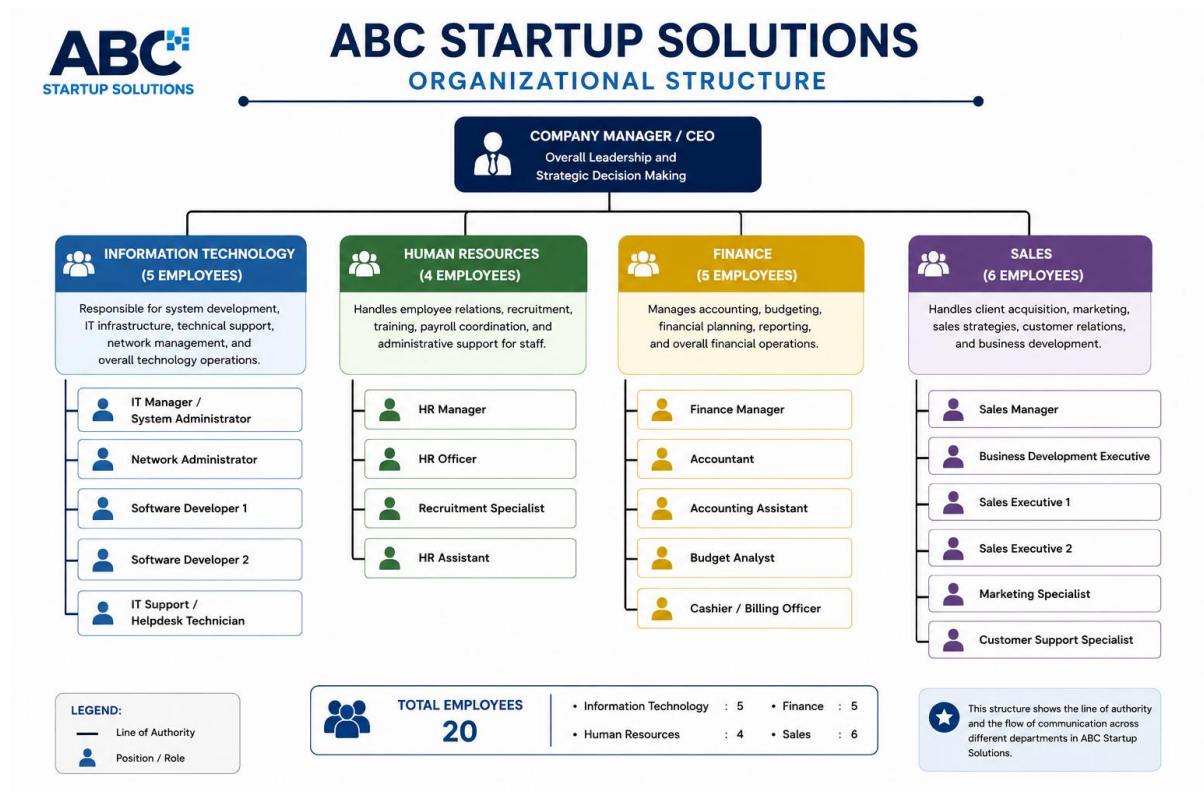
Company Mission

ABC Startup Solutions aims to develop quality software solutions that address the needs of its clients while maintaining professionalism, security, efficiency, and continuous innovation.

Office Location

3rd Floor, ABC Business Center, Santa Rosa, Laguna, Philippines

Organizational Structure



Employee Distribution

Department	Employees	Main Function
Information Technology	5	Software development, technical support, systems and network administration
Human Resources	4	Employee records, recruitment, payroll coordination
Finance	5	Accounting, budgeting, financial records
Sales	6	Client acquisition, marketing, sales and customer relations
TOTAL	20	

The employee distribution follows the scenario provided in the Week 2 module.

PART 2 — ENTERPRISE HARDWARE INVENTORY

The hardware plan is designed for 20 employees while providing additional shared infrastructure for server operations, networking, printing, storage, and backup.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	5	IT	Software development and technical administration
HW-002	Desktop Computer	4	HR	HR records and administrative tasks
HW-003	Desktop Computer	5	Finance	Accounting and financial operations
HW-004	Desktop Computer	6	Sales	Sales, customer management and documentation
HW-005	Laptop	5	IT	Development, meetings and mobile work
HW-006	Laptop	2	Management	Meetings and administrative work
HW-007	Server	1	IT	Centralized services and internal applications
HW-008	Router	1	Network	Internet routing and network management
HW-009	Managed Switch	1	Network	Connect wired devices and departments
HW-010	Printer	2	Shared	Printing business documents
HW-011	UPS	3	IT/Network	Power protection for server and network equipment
HW-012	Wireless Access Point	3	Network	Wireless connectivity throughout the office
HW-013	NAS Storage	1	IT	Centralized file storage and backup
HW-014	External Backup Drive	2	IT	Offline backup and disaster recovery
HW-015	Monitor	22	All Departments	Additional/display monitors for employees

Hardware Justification

20 desktop computers are provided because the company has 20 employees distributed across four departments. 5 laptops are assigned to IT personnel because their work may require mobility during meetings, testing, maintenance, and deployment. Additional laptops are provided for management.

A dedicated server is included for centralized internal services. A managed switch provides wired network connectivity, while wireless access points provide Wi-Fi coverage. NAS storage is used for centralized file storage and backup, while external backup drives provide an additional offline backup option.

UPS units are recommended to protect critical equipment from sudden power interruptions and allow systems to be shut down properly during extended outages.

The required hardware categories are based on the module's Enterprise Hardware Inventory requirements.

PART 3 — ENTERPRISE SOFTWARE INVENTORY

Software	Version	License	Purpose
Windows 11 Pro	Current supported version	Commercial	Operating system for employee computers
Ubuntu Server	Current LTS version	Open Source	Server operating system
Microsoft Office	Microsoft 365 / Current	Subscription	Documents, spreadsheets and presentations
Visual Studio Code	Current stable version	Free	Software development and code editing
Git	Current stable version	Open Source	Source code version control
GitHub Desktop	Current version	Free	Graphical Git and GitHub management
VirtualBox	Current version	Open Source	Virtual machine testing and development
Google Chrome	Current stable version	Free	Web browsing and web application testing
Microsoft Defender	Built-in/current	Included	Endpoint security and malware protection
AnyDesk	Current version	Commercial/Free options	Authorized remote technical support
7-Zip	Current stable version	Open Source	File compression and archive management

Software Justification

Windows 11 Pro will be used as the primary operating system for employee workstations because it provides a familiar business environment and supports the productivity and development applications required by the company.

Ubuntu Server will serve as the server operating system because it is an open-source Linux distribution suitable for server environments.

Microsoft Office will support documentation, spreadsheets, presentations, and administrative work.

Visual Studio Code will be used by the IT department for software development and code editing.

Git and **GitHub Desktop** will support source-code version control and collaboration among developers.

VirtualBox will allow IT personnel to create virtual machines for testing and learning without affecting production systems.

Google Chrome will be used for web browsing and testing web applications.

Microsoft Defender provides endpoint security and malware protection.

AnyDesk may be used for authorized remote technical assistance and troubleshooting.

7-Zip provides file compression and archive management.

The software list follows the minimum software requirements specified in the module.

PART 4 — ENTERPRISE NETWORK INVENTORY

Asset ID	Network Equipment	Quantity	Purpose
NET-001	ISP Modem	1	Provides connection from the internet service provider
NET-002	Router	1	Routes traffic between networks and the internet
NET-003	Firewall	1	Filters and controls network traffic
NET-004	Managed Switch	1	Connects wired computers and network devices
NET-005	Wireless Access Point	3	Provides wireless network access
NET-006	Patch Panel	1	Organizes and terminates network cabling
NET-007	CAT6 Cable	2 boxes	Wired network connections
NET-008	RJ45 Connectors	100 pcs	Cable termination and replacement

Network Design Justification

The proposed network uses a structured design where internet traffic passes through the ISP modem, router, and firewall before reaching the internal network. A managed switch connects wired devices such as computers, the server, and printers.

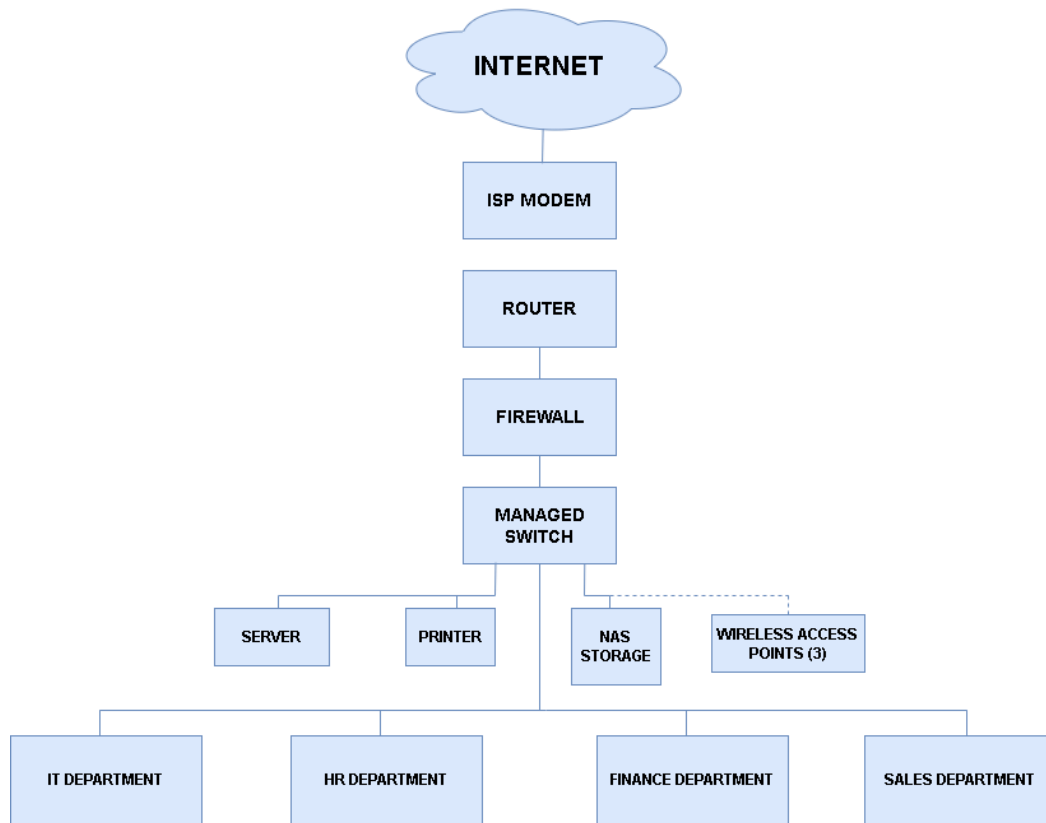
Three wireless access points are recommended to provide adequate coverage throughout the office floor. CAT6 cabling is selected for reliable wired connections and future network expansion.

The network inventory follows the required equipment listed in the Week 2 module.

PART 5 — ENTERPRISE NETWORK DIAGRAM

Proposed Network Topology

The network should be arranged in this logical order:



PART 6 — SYSTEM ADMINISTRATION ROLES

6.1 Helpdesk Technician

Responsibilities

A Helpdesk Technician provides first-level technical support to employees. They troubleshoot common hardware and software problems, assist users with account and application issues, document incidents, and escalate complex problems to specialized IT personnel.

Skills

- Troubleshooting
- Hardware and software knowledge
- Communication skills
- Customer service
- Problem-solving
- Basic networking
- Documentation

Common Tools

- Windows troubleshooting tools

- Remote support software
- Ticketing systems
- Microsoft Defender
- Device management tools
- Command-line utilities

Certifications

A beginner-level certification such as **CompTIA A+** can help demonstrate foundational IT support knowledge.

6.2 Network Administrator

Responsibilities

The Network Administrator manages and maintains the organization's network infrastructure. Responsibilities include configuring routers and switches, monitoring connectivity, maintaining network security, troubleshooting network problems, and managing network devices.

Skills

- TCP/IP
- Routing and switching
- Network troubleshooting
- Network security
- IP addressing
- Wireless networking
- Network monitoring

Common Tools

- Cisco IOS
- Packet Tracer
- Wireshark
- Ping
- Traceroute
- Network monitoring tools

Certifications

Cisco Certified Network Associate (CCNA) is a relevant certification because it covers networking fundamentals, network access, IP connectivity, IP services, security fundamentals, and automation/programmability.

Cisco also identifies CCNA and CCNP Enterprise among certifications relevant to network administrator career development.

6.3 Linux System Administrator

Responsibilities

A Linux System Administrator manages Linux-based servers and systems. Responsibilities include installing and updating software, managing users and permissions, configuring services, managing storage, monitoring systems, and maintaining system security.

Skills

- Linux command line
- File and directory management
- User and group management
- Networking
- Storage management
- Shell scripting
- Security administration
- System troubleshooting

Common Tools

- Bash
- SSH
- systemctl
- journalctl
- package managers
- Linux firewall tools
- Virtual machines

Certifications

Red Hat Certified System Administrator (RHCSA) is a relevant certification. Red Hat states that RHCSA validates core system administration skills including managing files, processes, storage, services, users, security, and basic containers.

6.4 Cloud Administrator

Responsibilities

A Cloud Administrator manages cloud-based computing resources and services. Responsibilities may include monitoring cloud resources, managing users and permissions, maintaining cloud security, managing storage and compute resources, monitoring costs, and assisting with cloud deployments.

Skills

- Cloud computing
- Networking
- Security
- Identity and access management
- Virtualization

- Storage
- Monitoring
- Basic automation

Common Tools

- AWS Management Console
- AWS CLI
- Cloud monitoring tools
- Identity and Access Management tools
- Virtual machines
- Cloud storage services

Certifications

The **AWS Certified Cloud Practitioner** is an appropriate foundational certification for cloud knowledge. AWS describes it as validating foundational knowledge of AWS Cloud, services, and terminology. Its coverage includes cloud concepts, security, cloud technology/services, and billing/pricing/support.

AWS also identifies compute, networking, databases, storage, security, and cloud deployment concepts within its certification coverage.

How the Four Professionals Work Together

The four professionals work together to maintain a reliable IT environment. The Helpdesk Technician handles common employee technical issues and escalates problems that require specialized knowledge. The Network Administrator manages connectivity, routers, switches, wireless access points, and network security. The Linux System Administrator maintains Linux servers and internal services. Meanwhile, the Cloud Administrator manages cloud resources and services. Their responsibilities overlap during troubleshooting and infrastructure changes, so communication and proper documentation are important to prevent disruptions and maintain business operations.

PART 7 — INFRASTRUCTURE RECOMMENDATIONS

Internet Provider

ABC Startup Solutions should subscribe to a business-grade fiber internet service with a reliable service-level agreement and sufficient bandwidth for 20 employees.

A target connection of approximately 500 Mbps to 1 Gbps is recommended depending on availability and budget.

Reason

A software development company regularly uses cloud services, source-code repositories, video conferencing, online collaboration tools, and web-based applications. Reliable internet connectivity is therefore critical to daily operations.

Server Specifications

Recommended initial server:

- 8-core processor
- 32 GB ECC RAM

- 2 × 1 TB SSD in RAID 1
- Additional storage for internal services
- Gigabit Ethernet
- UPS protection
- Ubuntu Server LTS

Reason

The server should have enough processing power and memory to support centralized internal services while allowing future expansion. RAID 1 provides redundancy for the primary storage.

Backup Strategy

The company should follow a 3-2-1 backup strategy:

- 3 copies of important data
- 2 different types of storage
- 1 copy stored separately/offline

Recommended backup locations:

1. Primary server
2. NAS storage
3. External backup drive stored separately

Important data should be backed up regularly and backup restoration should be tested periodically.

Security Recommendations

The company should implement:

- Firewall protection
 - Strong passwords
 - Multi-factor authentication where available
 - Regular software updates
 - User access controls
 - Separate user and administrator accounts
 - Secure Wi-Fi configuration
 - Regular backups
 - Security awareness training
 - Least-privilege access
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Antivirus

Microsoft Defender should be enabled and regularly updated on Windows workstations. Security monitoring should also be implemented to identify suspicious activity.

Password Policy

Employees should use strong and unique passwords. Passwords should not be shared between employees or reused across important systems.

The company should also implement:

- Multi-factor authentication
 - Account lockout/rate limiting
 - Separate administrator accounts
 - Secure password management
 - Immediate removal of access for employees who leave the company
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Expansion Plan

The infrastructure should be designed with future growth in mind.

If the company expands from 20 to 40 employees, management should consider:

- Additional computers
- Additional network switches
- Additional wireless access points
- Increased internet bandwidth
- Larger NAS capacity
- Additional server resources
- Additional network ports
- Improved cloud infrastructure

Planning for expansion prevents the company from having to redesign the entire infrastructure when employee numbers increase.

PART 8 — PERSONAL REFLECTION

This project helped me understand that system administration involves more than simply installing computers, configuring networks, or fixing technical problems. Before an IT infrastructure can be deployed, the needs of the organization must first be analyzed and documented. Through this activity, I learned how to identify the hardware, software, and networking requirements of a startup company based on the number of employees and the functions of each department.

The most challenging part of the project for me was designing the enterprise network topology. I had to think about how the Internet, modem, router, firewall, switch, server, printer, wireless access points, and different departments should be connected. I also needed to make sure that the diagram was logical and easy to understand. This task helped

me realize that network design requires careful planning because an incorrect connection or missing component can affect the entire organization.

I also learned the importance of creating inventories. A hardware or software inventory allows an organization to know what resources are available, where they are assigned, and what their purposes are. Proper documentation can make troubleshooting, maintenance, upgrades, and future expansion easier.

Planning is important before deployment because it helps prevent unnecessary expenses, compatibility problems, security issues, and poor resource allocation. Instead of purchasing equipment randomly, a system administrator should first understand the organization's requirements and create a plan that matches its current and future needs.

Overall, this project can help me become a better System Administrator by improving my technical planning, documentation, networking, and decision-making skills. It also taught me that a good system administrator needs both technical knowledge and the ability to communicate technical decisions clearly. The project gave me a better understanding of how different IT roles work together to support an organization's operations.
